Biology - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What controls the process of molting incrustacea?
   (a) Water temperature alone without hormonal influence
   (b) The age of the crustacean
   (c) Environmental factors
   (d) Social interactions with other crustaceans
   (e) Diet of the crustacean

2. A dominant trait has a fitness of 0.6. The frequency in a population of this trait is 1 in 8,000. What is the mutation rate?
   (a) 2.5×10^-5
   (a) 1.5×10^-5
   (a) 3.0×10^-5
   (a) 4.0×10^-5
   (a) 5.0×10^-5

3. Why is it incorrect to say that man evolved from monkeys? What did he possibly evolve from?
   (a) Man evolved directly from monkeys
   (b) Man evolved from a branch of primates that includes the modern lemur
   (c) Man evolved from Parapithecus
   (d) Man evolved from a direct line of increasingly intelligent monkeys
   (e) Man evolved from Neanderthals who were a separate species from monkeys

4. All of the following are examples of connective tissue EXCEPT
   (a) tendons
   (b) skin
   (c) muscle
   (d) blood
   (e) cartilage

5. Mutation of homeotic cluster genes often results in which of the following developmental defects in Drosophila?

(a) Complete absence of segmentation
(b) Hindrance in the development of the nervous system
(c) Inability to undergo metamorphosis
(d) Absence of a group of contiguous segments
(e) Tumor formation in imaginal discs

True / False (10 marks)

Answer True or False

6. True or False: Do general practice selection scores predict success at MRCGP?
7. True or False: Discharging patients earlier from Winnipeg hospitals: does it adversely affect quality of care?
8. True or False: Processing fluency effects: can the content and presentation of participant information sheets influence recruitment and participation for an antenatal intervention?
9. True or False: Blunt trauma in intoxicated patients: is computed tomography of the abdomen always necessary?
10. True or False: Resection of colorectal liver metastases after second-line chemotherapy: is it worthwhile?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of courtship behavior in the common tern, particularly the male's presentation of fish, and analyze its role in ensuring successful reproduction within the species.
12. Discuss the mechanisms by which the biological clock of fiddler crabs can be manipulated to reverse their color change pattern from dark during the day to pale at night.

End of Paper